

PocketLab Rev A Firmware Requirements

Revised ADS1220 / TMUX1109 Analog Architecture

Status: design note for schematic freeze and later firmware implementation. Revision date: 2026-08-18. This document reflects the revised unipolar resistance excitation plan while preserving the ADS1220 bipolar analog supply (+2.5 V / -2.5 V).

Core rule: voltage measurement remains bipolar-capable, but all valid resistance-mode probe signals are constrained to approximately 0 V through +2.5 V. The ADS1220 internal low-side PSW is not used.

1. Final ADS1220 analog pin assignment

ADS1220 input	Connection	Purpose
AIN0	VSENSE_BUF	Voltage measurement input
AIN1	GND	Voltage reference input and high-ohms negative ADC input
AIN2	OHMS_SENSE_LO	Resistance Rref/divider junction
AIN3	OHMS_SENSE_HI	Resistance probe-high node and IDAC1 destination
REFP0	ADC_REFP0	Ratiometric reference high
REFN0	GND	Ratiometric reference low

2. ADS1220 mode configuration

Mode	ADC MUX	MUX bits	Reference	PGA / IDAC
Voltage	AIN0 - AIN1	0000	Internal 2.048 V	Gain 1, PGA bypass, IDAC off
10k Rref	AIN3 - AIN2	0111	REFP0 / REFN0	IDAC on, PSW off
100k Rref	AIN3 - AIN2	0111	REFP0 / REFN0	IDAC on, PSW off
1M divider	AIN1 - AIN2	0011	+2V5A / GND	Gain 1, PGA bypass, IDAC off
10M divider	AIN1 - AIN2	0011	+2V5A / GND	Gain 1, PGA bypass, IDAC off

The ADS1220 datasheet defines MUX=0000 as AIN0-AIN1, 0011 as AIN1-AIN2, and 0111 as AIN3-AIN2. For high-ohms mode the conversion is intentionally negative because AIN1 is GND and AIN2 is the positive divider node.

3. External TMUX1109 functions

MUX_OHMS keeps the four range states. A1:A0 = 00/01/10/11 selects 10k/100k/1M/10M. Its A bank selects the Rref topology; its B bank selects REFP0. EN=0 must disconnect all channels.

A1 A0	Range	Rref connection	ADC_REFP0	IDAC / PSW
00	10k	10k to GND	OHMS_SENSE_LO	IDAC selected / PSW OFF
01	100k	100k to GND	OHMS_SENSE_LO	IDAC selected / PSW OFF
10	1M	1M to +2V5A	+2V5A	IDAC OFF / PSW OFF
11	10M	10M to +2V5A	+2V5A	IDAC OFF / PSW OFF

Repurposed MUX_ADC: DA connects to OHMS_SENSE_HI. S1A and S2A are NC/open. S3A and S4A connect to GND. The B bank is unused. Prefer MUX_ADC_EN=0 in voltage and low/mid-ohms modes, and MUX_ADC_EN=1 only in the 1M/10M divider modes. TMUX1109 EN=0 turns all channels off.

4. Resistance-mode electrical behavior

Low/mid ohms (10k and 100k Rref): IDAC1 is routed to AIN3/OHMS_SENSE_HI. Current flows through Rx and the selected Rref to GND. The ADC measures AIN3-AIN2 while REFP0 is tied to OHMS_SENSE_LO and REFN0 to GND. This remains ratiometric because the same excitation current flows through Rx and Rref.

High ohms (1M and 10M Rref): the selected Rref is connected from +2V5A to OHMS_SENSE_LO. The repurposed TMUX1109 grounds OHMS_SENSE_HI. Rx therefore sits between OHMS_SENSE_LO and GND. REFP0 is +2V5A and REFN0 is GND. The ADC measures AIN1-AIN2 = GND - OHMS_SENSE_LO.

High-ohms conversion: sign-extend the 24-bit conversion to a signed integer. For a valid divider reading the code should be negative. Define $k = -\text{code} / 2^{23}$ (gain 1), then compute $R_x = R_{\text{ref}} * k / (1 - k)$. Reject $k \leq 0$, $k \geq 1$, unexpected positive code, or values too close to either endpoint.

5. IDAC requirements and compliance

The ADS1220 IDAC output pin must remain at or below approximately $AVDD - 0.9\text{ V}$ for specified IDAC accuracy. With $AVDD = +2.5\text{ V}$, the AIN3/OHMS_SENSE_HI target is therefore $\leq$ about $+1.6\text{ V}$ while an IDAC is active.

Example setting	Approx. total R at 1.6 V compliance	Approx. Rx headroom after Rref
100 μA , 10k Rref	16k ohm	6k ohm
50 μA , 10k Rref	32k ohm	22k ohm
10 μA , 100k Rref	160k ohm	60k ohm

Do not freeze autorange thresholds from these estimates alone. Bench characterization should choose the current/PGA combination that keeps the IDAC in compliance while maintaining adequate ADC utilization. The earlier 100 μA / 10k and 10 μA / 100k values remain useful starting points, not unconditional full-range settings.

6. Required mode-transition sequence

1. Enter safe state: IDAC off, PSW off, MUX_ADC_EN=0, MUX_OHMS_EN=0, ohms relays open.
2. If entering resistance mode, first verify the DUT is not externally energized and no protection fault is present.
3. Program A1/A0 for the desired range while the muxes are disabled.
4. Configure ADS1220 input MUX, reference source, gain, PGA bypass, and IDAC current.
5. For low/mid ohms: enable MUX_OHMS, route IDAC1 to AIN3 only after programming the IDAC current, then wait at least 200 μs for IDAC startup.
6. For high ohms: enable MUX_OHMS, then enable MUX_ADC ground-switch state; IDAC remains off.
7. Close the ohms relays only in the intended safe sequence; allow relay and analog settling.
8. Start/synchronize conversion, discard initial unsettled conversion(s) as required by bench characterization, then accept data.
9. On fault, reset, brownout, overrange, invalid ratio, or mode change: IDAC off first, disable muxes, then open ohms relays.

7. Provisional firmware configuration by mode

Item	Voltage	10k/100k ohms	1M/10M ohms
OHMS relays	Open	Closed after safety check	Closed after safety check
MUX_OHMS	Disabled	Enabled; range selected	Enabled; range selected
MUX_ADC	Disabled	Disabled	Enabled; S3/S4 -> GND
ADS MUX	0000	0111	0011
VREF bits	00 internal	01 external REFP0/REFN0	01 external REFP0/REFN0
PSW	0	0	0
IDAC	Off	Selected by range/compliance	Off
IDAC1 route	Disabled	AIN3 (I1MUX=100)	Disabled
PGA	Bypass initially	Range-dependent	Bypass
Expected sign	+/- voltage	Positive Rx drop	Negative code; negate in FW

8. Calibration and diagnostic requirements

- Store calibrated Rref values per board for 10k, 100k, 1M, and 10M ranges.
- Low-ohm calibration must include probe/lead resistance, reed-relay contact resistance, R_PROT_HI/LO, and any series switch resistance in the active path.
- Characterize zero offset separately for each gain/current/range combination; do not apply one universal short correction.
- High-ohms calibration must characterize leakage after PCB cleaning and under representative humidity/temperature conditions.
- Log the selected range, ADC register state, raw signed code, calculated ratio k, calibrated Rref, and protection/fault status during development.
- Autorange should prefer the middle of the useful ADC ratio span and avoid ratios close to 0 or 1 when an overlapping range exists.

9. Schematic items to correct before freeze

- **MUX_ADC wiring:** S1A/S2A must be NC; S3A/S4A must be GND. In the supplied screenshot S1A/S2A are presently tied to GND, which is reversed for the intended range coding.
- **MUX_OHMS truth-table note:** remove the old high-ohms 'PSW CLOSE' and 'AVDD-AVSS reference' text. High-ohms now uses PSW OFF and external +2V5A/GND reference.
- **ADC firmware note:** remove the old statement that 1M/10M uses the analog-supply reference and PSW. Replace it with the revised note provided with this document.
- **Protection stage:** TMUX7462F threshold/supply compatibility remains a separate open item. Do not declare the schematic frozen until that protection issue is resolved.
- **Probe loading check:** review the permanently connected voltage-divider path during resistance mode. If it remains across the probes, either isolate it in hardware or explicitly include its parallel loading in the resistance model/calibration, especially on the 1M/10M ranges.

10. References

- Texas Instruments, ADS1220 datasheet, Rev. D, May 2026 (SBAS501D).
- Texas Instruments, TMUX1109 datasheet, Rev. A, February 2024 (SCDS406A).
- PocketLab DMM Resistance Measurement Proof of Concept engineering validation report, August 8, 2026.